

LRSC Odd-Ring Spectral Rank Theorem

Version 1.2.1 — Proof Clarification and Verification Supplement

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Abstract

This supplement makes the finite Fourier calculation and channel construction in the frozen LRSC odd-ring theorem release explicit. It fixes the DFT convention, defines every channel entry, evaluates the finite cosine sums by a stated progression identity, and maps each hypothesis to the rank conclusion. It also reports a separately written numerical audit, clearly distinguished from the proof. Version 1.2 and its OSF registration (DOI [10.17605/OSF.IO/NM5BW](https://doi.org/10.17605/OSF.IO/NM5BW)) remain frozen and unchanged. This document is explanatory and supplementary; it does not replace, amend, or silently revise the registered theorem.

1 Purpose, scope, and provenance

The result addressed here is a theorem about a specified single-step operator on an odd cyclic lattice. The task of this note is narrower than a new result: it spells out the intermediate algebra and linear-algebra objects so that a reader can check the proof without inferring conventions. The theorem statement and scope remain those of v1.2.

The frozen source release is [LRSC Odd-Ring Spectral Rank Theorem v1.2](#); its OSF registration is [NM5BW](#), DOI [10.17605/OSF.IO/NM5BW](https://doi.org/10.17605/OSF.IO/NM5BW). The companion verifier in that release is corroborative, not a premise of the analytic proof. This supplement is not itself represented as DOI-registered.

2 Definitions and theorem

Let $M \geq 9$ be odd, $k = (M - 1)/2$, and $\alpha = \pi/M$. Indices lie in $\mathbb{Z}_M$. We use the centered representatives $t = -k, \dots, k$ when describing the signal and the representatives $i = 0, \dots, k$ for reflection orbits $\{i, -i\}$ (with $\{0\}$ the singleton orbit).

Define

$$X_t = 1 + A \cos\left(\frac{2\pi kt}{M}\right), \quad A > 0, \quad j_t = X_t - \min(X_{t-1}, X_t, X_{t+1}).$$

Let $m_i \in \{0, 1\}$ obey $m_i = m_{-i}$. Assume at least one reflection orbit is inactive, and let p be the number of active reflection orbits.

For any $x \in \mathbb{C}^M$, fix the DFT convention

$$\hat{x}_q = \sum_{t=0}^{M-1} x_t e^{-2\pi i q t / M}, \quad x_t = \frac{1}{M} \sum_{q=0}^{M-1} \hat{x}_q e^{2\pi i q t / M}.$$

Thus $\hat{j}_q/M$ is the normalized Fourier coefficient. Since j is real and even, $\hat{j}_q = \hat{j}_{M-q} \in \mathbb{R}$.

Define the real conjugate-balanced channels, for $i \in \mathbb{Z}_M$, by

$$\begin{aligned} w_0(i) &= m_i \frac{\hat{j}_0}{M}, \\ w_r(i) &= m_i \left(\frac{\hat{j}_r}{M} e^{2\pi i r i / M} + \frac{\hat{j}_{M-r}}{M} e^{-2\pi i r i / M} \right) = 2m_i \frac{\hat{j}_r}{M} \cos\left(\frac{2\pi r i}{M}\right), \quad 1 \leq r \leq k. \end{aligned}$$

Here $W \in \mathbb{R}^{M \times (k+1)}$ has entries $W_{i,r} = w_r(i)$, with columns indexed $r = 0, \dots, k$. Including the common factor $1/M$ in each Fourier coefficient fixes the normalization; omitting it would only rescale all columns and would not change rank.

Let F be the DFT matrix under the convention above and define the real circulant stencil

$$S = F^{-1} \text{diag}(s_0, \dots, s_{M-1}) F, \quad s_q = \cos\left(\frac{2\pi q}{M}\right) - 1.$$

The post-stencil matrix is $V = SW$.

Theorem (v1.2, unchanged). Under these assumptions,

$\text{rank}(V) = p.$

3 Exact shedding field and explicit finite Fourier sum

The centered carrier identity is

$$\cos\left(\frac{2\pi k t}{M}\right) = \cos((\pi - \alpha)t) = (-1)^t \cos(\alpha t),$$

so $X_t = 1 + A(-1)^t \cos(\alpha t)$. For odd $|t|$, the center value is the smaller of the three neighboring carrier values, giving $j_t = 0$. At zero, $j_0 = A(1 + \cos \alpha)$. At nonzero even sites, the smaller neighbor is the adjacent odd site in the direction of the center, hence

$$j_t = A\{\cos(|t|\alpha) + \cos((|t| - 1)\alpha)\}, \quad 0 < |t| \leq k, \quad |t| \text{ even}.$$

This also shows directly that j is real and reflection-even. Put $L = \lfloor (M - 1)/4 \rfloor$. Then the nonzero positive centered even representatives are exactly $2, 4, \dots, 2L$, and

$$\frac{\hat{j}_r}{M} = \frac{1}{M} \left[j_0 + 2 \sum_{u=1}^L j_{2u} \cos(4ru\alpha) \right], \quad 0 \leq r < k. \quad (1)$$

The two adjacent cosines in j_{2u} combine exactly:

$$j_{2u} = A\{\cos(2u\alpha) + \cos((2u - 1)\alpha)\} = 2A \cos(\alpha/2) \cos((4u - 1)\alpha/2). \quad (2)$$

Also $j_0 = 2A \cos^2(\alpha/2)$. Substituting (2) in (1) gives

$$\frac{\hat{j}_r}{M} = \frac{2A \cos(\alpha/2)}{M} B_r, \quad B_r = \cos(\alpha/2) + 2 \sum_{u=1}^L \cos((4u - 1)\alpha/2) \cos(4ru\alpha). \quad (3)$$

Applying $2 \cos a \cos b = \cos(a+b) + \cos(a-b)$ to each summand yields the two arithmetic progressions

$$B_r = \cos(\alpha/2) + \sum_{u=1}^L \cos\left(\left[2(2r+1)u - \frac{1}{2}\right] \alpha\right) + \sum_{u=1}^L \cos\left(\left[2(2r-1)u + \frac{1}{2}\right] \alpha\right). \quad (4)$$

For completeness, if $d \notin 2\pi\mathbb{Z}$, summing the geometric progression of $e^{i(ud+\beta)}$ and taking its real part gives

$$\sum_{u=1}^L \cos(ud + \beta) = \frac{\sin(Ld/2)}{\sin(d/2)} \cos\left(\beta + \frac{(L+1)d}{2}\right). \quad (5)$$

Indeed, $\sum_{u=1}^L e^{i(ud+\beta)} = e^{i(\beta+(L+1)d/2)} \sin(Ld/2)/\sin(d/2)$; (5) follows by taking real parts. Applying (5) separately to the two sums in (4), with $(d, \beta) = (2(2r+1)\alpha, -\alpha/2)$ and $(2(2r-1)\alpha, +\alpha/2)$, gives the explicit evaluated finite sum

$$\begin{aligned} B_r = & \cos(\alpha/2) + \frac{\sin(L(2r+1)\alpha)}{\sin((2r+1)\alpha)} \cos((L+1)(2r+1)\alpha - \alpha/2) \\ & + \frac{\sin(L(2r-1)\alpha)}{\sin((2r-1)\alpha)} \cos((L+1)(2r-1)\alpha + \alpha/2). \end{aligned} \quad (6)$$

No unevaluated sum remains in (6).

To reduce (6) to the compact residue formulas, the only residue-specific substitutions are

M	$L\alpha$	$(L+1)\alpha$
$4L+3$	$\pi/4 - 3\alpha/4$	$\pi/4 + \alpha/4$
$4L+1$	$\pi/4 - \alpha/4$	$\pi/4 + 3\alpha/4$

(7)

and the product-to-sum identities $2 \sin x \cos y = \sin(x+y) + \sin(x-y)$ and $\sin(u+v) \sin(u-v) = \sin^2 u - \sin^2 v$. To make the common-denominator reduction explicit, put $D_r = \sin((2r-1)\alpha) \sin((2r+1)\alpha)$. Substitution of the first row of (7) into (6), expansion of the two products, and collection over D_r gives

$$D_r B_r = -\frac{\sin(2\alpha)}{2} [(-1)^r \cos(r\alpha) + \sin(\alpha/2)], \quad M = 4L+3. \quad (8)$$

Using the second row gives

$$D_r B_r = -\sin \alpha [(-1)^r \cos(r\alpha) \cos(2r\alpha) + \cos \alpha \sin(\alpha/2)], \quad M = 4L+1. \quad (9)$$

Here is the product-expansion step used in that collection. Write $u = (2r-1)\alpha$ and $v = (2r+1)\alpha$. In (6), use $2 \sin x \cos y = \sin(x+y) + \sin(x-y)$ on each numerator product. Before collecting terms, the result is

$$\begin{aligned} 2D_r B_r = & 2D_r \cos(\alpha/2) + \sin u \{\sin((2L+1)v - \alpha/2) - \sin(v - \alpha/2)\} \\ & + \sin v \{\sin((2L+1)u + \alpha/2) - \sin(u + \alpha/2)\}. \end{aligned} \quad (9a)$$

For $M = 4L+3$, $(2L+1)\alpha = \pi/2 - \alpha/2$, so the first sine in the first brace is $(-1)^r \cos((r+1)\alpha)$ and the first sine in the second brace is $(-1)^{r-1} \cos((r-1)\alpha)$. For $M = 4L+1$, $(2L+1)\alpha = \pi/2 + \alpha/2$, and those two sines become $(-1)^r \cos(r\alpha)$ and $(-1)^{r-1} \cos(r\alpha)$. The remaining four sine terms have arguments $2r\alpha \pm \alpha/2$. Substituting these values into (9a) and using $\sin(u+v) \sin(u-v) = \sin^2 u - \sin^2 v$ gives exactly (8) and (9), respectively. Equations (6)–(9) therefore show the finite progressions, their exact summation, and the common-denominator reduction. In particular,

$$B_r = -\frac{\sin(2\alpha)[(-1)^r \cos(r\alpha) + \sin(\alpha/2)]}{2D_r} \quad (M = 4L+3), \quad (10)$$

and

$$B_r = -\frac{\sin \alpha [(-1)^r \cos(r\alpha) \cos(2r\alpha) + \cos \alpha \sin(\alpha/2)]}{D_r} \quad (M = 4L + 1). \quad (11)$$

Thus the nonvanishing proof below rests on explicit finite expressions rather than an asserted summation step.

4 Non-vanishing for every required mode

For $0 \leq r < k$, neither factor in D_r vanishes: $2r - 1$ and $2r + 1$ are nonzero integers of magnitude less than M , so their products with $\alpha = \pi/M$ are not integer multiples of π . Also $A > 0$, $\cos(\alpha/2) > 0$, and the sine factors outside the brackets in (10)–(11) are positive. It remains to show that each bracket is nonzero.

4.1 $M = 4L + 3$

For even r , the bracket in (10) is $\cos(r\alpha) + \sin(\alpha/2) > 0$. For odd r ,

$$r\alpha \leq (k - 1)\alpha = \pi/2 - 3\alpha/2, \quad \text{so} \quad \cos(r\alpha) \geq \sin(3\alpha/2) > \sin(\alpha/2).$$

Consequently $-\cos(r\alpha) + \sin(\alpha/2) < 0$. The bracket cannot vanish.

4.2 $M = 4L + 1$

Here $k = 2L$, and a zero in (11) would imply

$$|\cos(r\alpha) \cos(2r\alpha)| = c, \quad c = \cos \alpha \sin(\alpha/2). \quad (12)$$

Because $M \geq 9$, $\alpha \leq \pi/9$ (in this residue class the first admissible M is 13).

For $0 \leq r \leq L - 1$, let $f(x) = \cos x \cos 2x$. On $[0, \pi/4)$,

$$f'(x) = -\sin x \{\cos 2x + 4 \cos x\} < 0.$$

Since $(L - 1)\alpha = \pi/4 - 5\alpha/4$,

$$f(r\alpha) \geq \cos(\pi/4 - 5\alpha/4) \sin(5\alpha/2) > \frac{5\alpha}{\pi\sqrt{2}} > \frac{\alpha}{2} > c. \quad (13)$$

Here the first strict bound uses $\cos(\pi/4 - 5\alpha/4) > 1/\sqrt{2}$ and $\sin(5\alpha/2) \geq 5\alpha/\pi$; the last uses $\cos \alpha < 1$ and $\sin(\alpha/2) < \alpha/2$.

At $r = L$, $L\alpha = \pi/4 - \alpha/4$ and $2L\alpha = \pi/2 - \alpha/2$, whence

$$|\cos(L\alpha) \cos(2L\alpha)| = \cos(L\alpha) \sin(\alpha/2) < \cos \alpha \sin(\alpha/2) = c, \quad (14)$$

because $L\alpha > \alpha$ and cosine decreases on $[0, \pi]$.

For $L + 1 \leq r \leq 2L - 1$, set $s = 2L - r$ and $t = (s + 1/2)\alpha$. Then $1 \leq s \leq L - 1$, so

$$3\alpha/2 \leq t \leq \pi/4 - 3\alpha/4, \quad r\alpha = \pi/2 - t, \quad |\cos(r\alpha) \cos(2r\alpha)| = h(t) := \sin t \cos 2t.$$

On this interval,

$$h'(t) = \cos t(1 - 6 \sin^2 t),$$

which changes sign at most once, from positive to negative. Thus any interior critical point is a maximum and the minimum is at an endpoint. At the lower endpoint, using $\sin(3\alpha/2)/\sin(\alpha/2) = 1 + 2\cos\alpha$,

$$\frac{h(3\alpha/2)}{c} = \frac{(1 + 2\cos\alpha)\cos 3\alpha}{\cos\alpha} \geq \frac{1 + 2\cos\alpha}{2\cos\alpha} > 1. \quad (15)$$

At the upper endpoint, the same identity and $\sin(\pi/4 - 3\alpha/4) \geq 1/2$ give

$$\frac{h(\pi/4 - 3\alpha/4)}{c} = \sin(\pi/4 - 3\alpha/4) \frac{1 + 2\cos\alpha}{\cos\alpha} \geq \frac{1 + 2\cos\alpha}{2\cos\alpha} > 1. \quad (16)$$

The inequalities use $\alpha \leq \pi/9$, so $\cos 3\alpha \geq 1/2$. Hence $h(t) > c$ throughout this range. Together (13)–(16) show that equality in (12) is impossible. Therefore $\hat{j}_r \neq 0$ for every $0 \leq r < k$ in both odd residue classes.

5 Proof map: hypotheses to rank

The rank argument can be read as four implications.

1. **Odd-ring signal** $\rightarrow$ **nonzero low modes**. The exact shedding calculation and (10)–(16) give $\hat{j}_r \neq 0$ for $0 \leq r < k$.
2. **Inactive orbit** $\rightarrow p \leq k$. An odd ring has $k + 1$ reflection orbits. At least one is inactive, so the active count satisfies $p \leq k$. Thus all first p Fourier scales are nonzero.
3. **Distinct active orbits** $\rightarrow$ **pre-stencil rank** p . If $p = 0$, then $W = 0$ and the conclusion is immediate. For $p \geq 1$, choose representatives $i_s \in \{0, \dots, k\}$ for the active orbits, $s = 0, \dots, p - 1$, and put $x_s = \cos(2\pi i_s/M)$. Since cosine is strictly decreasing on $[0, \pi]$, these x_s are distinct. The $p \times p$ submatrix using these rows and channel columns $r = 0, \dots, p - 1$ is

$$W_{i_s, r} = c_r T_r(x_s), \quad c_0 = \hat{j}_0/M, \quad c_r = 2\hat{j}_r/M \quad (r \geq 1),$$

where T_r is the Chebyshev polynomial of degree r . Every $c_r \neq 0$. If a_r is the leading coefficient of T_r ($a_0 = 1$, $a_r = 2^{r-1}$ for $r \geq 1$), polynomial evaluation gives

$$\det[T_r(x_s)]_{s, r=0}^{p-1} = \left(\prod_{r=0}^{p-1} a_r \right) \left(\prod_{0 \leq u < v < p} (x_v - x_u) \right) \neq 0.$$

Hence $\text{rank}(W) \geq p$. Conversely, each column is mask-supported and reflection-even, so its values are determined by at most the p active orbit values; therefore $\text{rank}(W) \leq p$. Thus $\text{rank}(W) = p$.

4. **Inactive site** $\rightarrow$ **stencil preserves that rank**. The multiplier s_q vanishes exactly at $q = 0$, so $\ker S = \text{span}\{\mathbf{1}\}$. Every vector in $\text{span}(W)$ is zero on every inactive site. Since at least one such site exists, $\ker S \cap \text{span}(W) = \{0\}$. The restriction of S to this p -dimensional space is injective; consequently $\text{rank}(V) = \text{rank}(SW) = p$.

This explains why both parts of the mask assumption matter. Reflection symmetry makes the active masked space have dimension at most p ; an inactive orbit both forces $p \leq k$ for the non-vanishing argument and rules out a nonzero constant vector in the stencil kernel. The all-active-mask case is outside this theorem.

6 Independent numerical checks (corroboration only)

A separately written check script accompanies this supplement. It constructs j directly from the local-minimum definition, computes its DFT independently, evaluates the finite sum in (1) directly,

and compares both calculations with the compact formulas (10)–(11). It separately builds the channels and applies S by FFT for rank checks. It does not call the v1.2 verifier. The script uses $A = 0.5$ for numerical checks; the analytic proof covers every $A > 0$.

Independent check	Outcome
Odd rings checked, $M = 9, 11, \dots, 401$	197
Fourier coefficients checked, all $0 \leq r < k$	20,094
Largest absolute gap: direct finite sum vs. compact formula	$< 2 \times 10^{-14}$
Largest absolute gap: direct DFT vs. compact formula	$< 2 \times 10^{-14}$
Rank cases: all $p = 1, \dots, k$ for $M \in \{9, 11, 13, 17, 25\}$	35/35 pass
In each rank case, both W and SW have numerical rank p	35/35 pass

The independent calculation is finite, floating-point evidence. It checks the algebra and the defined matrix construction on selected inputs; it is not used to infer the universal result. The analytic proof above establishes the statement for all odd $M \geq 9$ under the stated mask assumptions.

7 Limitations and release boundary

The theorem is a structural rank result for the defined single-step cyclic operator. It does not establish a universal fixed-unit compression number K_τ , an all-active-mask conclusion, multi-step dynamics, a microscopic physical law, or the separate entropy-to-horizon bridge. Numerical checks do not enlarge these claims.

This supplement clarifies v1.2; it does not alter the v1.2 PDF, source files, manifest, OSF registration, or DOI. Any later deposit should be identified as a separate supplement and should retain a verifiable provenance link to the frozen theorem release.